



Inspiring Excellence

Department of Electrical and Electronic Engineering
EEE499P :Final Year Design Project (FYDP)
Weekly Update

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Week 4 Update: Multiple Design Approach

Subsystems:

1. **Processing Unit:**
2. **Communication Unit:**
3. **Current Sensor Unit:**
4. **Power Unit:**
5. **Base Station Unit:**
6. **Indicator Unit:**

Sub Functions	Means 1	Means 2	Means 3
Processing Unit	Microcontroller (for e.g, Arduino series, STM32 series)	Single-Board Computer (e.g. Raspberry Pi)	FPGA-based accelerators
Communication Unit	Cellular IOT For eg. LTE-M, 3G/4G SimCard	LoRA WAN (Long Range WAN)	Satellite
Current Sensor Unit	Rogowski coil	Hall current sensor	Shunt Resistor
Power Unit	CR2032 Coin cell	Solar Rechargeable Battery	
Base Station Unit	GSM/GPRS Network systems	4G/LTE Cellular network	LoRaWAN Gateway systems
Indicator Unit	LED	Sound/alarm	

Detailed Descriptions:

1. Processing Unit-

- A) Microcontrollers are essential for Fault Passage Indicators (FPIs), offering cost-efficient computation for fault detection. Arduino series (Uno, Nano etc) are great for prototyping due to their ease of use and community support. For industrial FPIs, STM32 series microcontrollers, particularly STM32F4 and STM32H7, are much more powerful. These ARM Cortex-M processors support real-time signal processing with hardware floating-point units, consume low power suited for remote use, and include ADCs and communication interfaces for direct sensor connections.
- B) Single-Board Computers significantly improve processing power, facilitating advanced fault diagnosis. The Raspberry Pi series, equipped with multi-core ARM processors and ample RAM, enables convolutional neural networks for precise fault classification. Unlike microcontrollers, SBCs support full operating systems, implementing the time-domain analysis of voltage and current signals without requiring FFT or wavelet transform preprocessing.
- C) Integrating an FPGA-based accelerator with neural network algorithms allows for real-time, efficient processing of electrical fault data at the edge. FPGAs offer low latency and low power consumption, ideal for remote environments. Techniques like model quantization and efficient memory access enhance local processing of sensor data, enabling quicker fault detection and response. This improves the reliability of smart grid systems while keeping operational costs low and reducing dependence on cloud connectivity.

2. Communication Unit-

- A) The cellular network can be used to send the signals to the central servers as well as to local operators who are on-field duty. It can use LTE-M modules or cellular modules that are inbuilt on Arduinos/ESPs. A sim card could also be used to send the messages to the operators on duty. This would require an additional sim plan for each FPI. In case of large distances one drawback is the absence of cellular towers could render it out of use.
- B) LoRA WAN. These modules could be used to transmit the signals. Different modules have different amounts of ranges (starting from a few metres to as long as 30km). This could be well used for our prototype and the larger range modules be used in real life. We need to calibrate the relevant communication protocol with it.

- C) Satellite. Here the idea is to equip the FPI with a satellite modem that could communicate with Low Earth Orbit Satellites and send the signals directly to the central server. This perhaps is also not the most desirable approach to fit in the prototype.

3.Current Sensor Unit:

The fault current specifications are:

Voltage levels	Fault Type	Normal Current Flow(1 phase)	Fault Current Flow (Taken from the switchgear ratings)
11kv	Line to Ground	50-400	470A
20kv(Data From A Reference Paper)	Line to Ground	80A	170A
33kv	Line to Ground	600-1200A	1790A

The Methodologies we can use here are:

- 1.Rogowski Coil: Available sensors and ratings: FR200RD Current Sensor 0A~10kA

2.Hall current Sensor

3.Shunt current Sensor:

Calculations for 200A/75mv Shunt

$$R_{shunt} = V_{rated} / I_{rated}$$

$$R_{shunt} = 0.075 / 200 = 0.000375 \text{ ohm}$$

Then At 10mA , Voltage drop will be=0.00000375V

Then At 100A , Voltage drop will be=37.5mV

4. Power Unit:

- A. The power unit supplies stable and uninterrupted operating power to the Advanced Overhead Field Passage Indicator system, ensuring reliable current monitoring and fault indication. It typically consists of a rectified and regulated **power supply stage that converts the overhead line induction or auxiliary input to the required DC levels. The system operates at 3.3V DC for the microcontroller**, 5V DC for sensor interfacing and signal conditioning circuits, and includes a CR2032 3V coin-cell battery dedicated to powering the Real-Time Clock (RTC) module for continuous timestamp tracking even during power interruptions.

- B. A compact stand-alone solar PV system can efficiently power pole- or line-mounted FPIs. A small **solar panel charges a Li-ion or LiFePO₄ battery through an MPPT charge controller and BMS**, providing stable power to the RL78 MCU, ADC, and sensors. A supercapacitor handles short high-current loads for LEDs or radio bursts, while duty-cycling and sleep modes keep average power below the solar input. LiFePO₄ batteries offer better temperature range and cycle life, while 18650 Li-ion cells suit space-constrained designs. Use an MPPT IC (e.g., from e-peas or TI) with a 1–10 W panel and size the PV and battery for at least 3–5 days of autonomy. A low-noise 3.3 V

regulator ensures clean power for the ADC, with filtering and transient protection. Include temperature-compensated charging and a coin-cell RTC backup for reliable operation during low-sun periods.

5. Base Station Unit-

- A) **GSM/GPRS communication is an effective low-bandwidth connectivity solution for FPIs**, suitable for basic fault notifications in remote areas. Utilizing existing cellular infrastructure reduces deployment costs, though its limited data rate restricts it to simple alerts rather than complex data. Through ThingSpeak, an IoT interface, these alerts can be visualized. A local edge server preprocesses notifications and forwards them to a database like MongoDB, ensuring smooth flow during network disruptions and when affordability and coverage are prioritized.
- B) **The 4G LTE protocol provides high-speed, IP-based connectivity**, ideal for data-driven FPI systems. It enables streaming of fault parameters, waveforms, and geographical data to cloud servers. LTE supports IoT protocols like MQTT and WebSockets for low-latency communication. With Blynk dashboards for live data visualization and MongoDB for storage, it surpasses GSM/GPRS in responsiveness, making it perfect for smart grid applications needing real-time operator engagement.
- C) In contrast, **LoRaWAN systems prioritize long-range, low-power, and low data-driven** communication, which is good for remote FPI deployments lacking reliable cellular connectivity. Operating in the sub-GHz band, it provides coverage up to 15 km in rural areas with low power usage. Multiple FPIs connect to a single LoRa gateway, lowering infrastructure costs. While it has limited data capacity, LoRaWAN can transmit alerts and status updates to edge servers, which relay data to dashboards like ThingSpeak or Blynk, while also archiving in MongoDB for analytics.

6. Indicator Unit:

LED:

The light blinks in different colors or patterns to signal the type of fault and also to correctly identify the position.(LED RED CLEAR 2SMD)

Sound/ Alarm:

The sound can be helpful in order to find the fault location by the surveillance team while in field identification.

Simulation Ideas:

For the simulation we can use Powerfactory which can allow us to measure and develop an algorithm for finding fault current at any distance while keeping the impedance of the transmission line in consideration.

